

PROJECT REPORT

Sheep Age Detection Model

Using Deep Learning and Transfer Learning

Title: Sheep Age Detection Project

Subtitle: Deep Learning-based Sheep Teeth Age Classification

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Abstract

In the domain of **Precision Livestock Farming (PLF)**, accurate age estimation of animals plays a vital role, particularly in religious and commercial practices such as **Qurbani**, where age eligibility is a strict requirement. Traditionally, sheep age determination is performed through **manual inspection of dentition (teeth)**, which requires expert knowledge and is often time-consuming, subjective, and prone to human error.

This project presents the development of a **Sheep Age Detection System** using **Deep Learning and Transfer Learning techniques**, designed to automate the process of age estimation from sheep teeth images. The system classifies sheep into **four distinct age categories** relevant to Qurbani eligibility:

- **3–12 months**
- **1–1.5 years**
- **1.5–2 years**
- **2–3 years (Qurbani eligible)**

Due to the limited availability of labeled veterinary datasets, a relatively small dataset consisting of approximately **50–60 images per class** was utilized. To improve model robustness and reduce overfitting, **data augmentation techniques** such as rotation, flipping, and rescaling were applied during training.

The proposed system is based on a **pretrained Convolutional Neural Network (CNN)** architecture using **transfer learning**, followed by **Global Average Pooling** and **fully connected layers** for classification. The model was trained for **20 epochs** using the **Google Colab** environment and achieved a validation accuracy of approximately **77%**, which is reasonable given the dataset size and complexity.

After training, the model was successfully converted into **TensorFlow Lite (.tflite)** format and **integrated into a Flutter-based mobile application**, enabling on-device inference and real-time age prediction. This implementation demonstrates the practical applicability of the system for farmers and livestock handlers, allowing efficient verification of sheep age for Qurbani purposes.

1. Introduction

1.1 Background

Artificial Intelligence (AI) and Computer Vision are rapidly transforming multiple real-world domains, including agriculture and livestock management. One particularly important application in this field is the determination of sheep age for **Qurbani eligibility**. In Islamic practices, animals must meet specific age criteria to be considered valid for Qurbani, making accurate age identification both religiously and economically significant.

Traditionally, the age of a sheep is determined through manual examination of its teeth, as milk teeth are gradually replaced by permanent incisors at known stages of growth. While this method is widely accepted, it requires expert knowledge and physical handling of the animal. For farmers, traders, and buyers—especially in crowded livestock markets—this process can be difficult, time-consuming, and prone to human error.

With recent advancements in **Deep Learning and Computer Vision**, it has become possible to automate such visual inspection tasks. By analyzing images of sheep teeth, AI-based systems can provide a fast, reliable, and non-invasive solution for estimating age, making them highly suitable for Qurbani verification and livestock management.

1.2 Project Scope

This project focuses on the development of an **automated Sheep Age Detection System** using Deep Learning techniques. The system takes an image of a sheep's teeth as input and predicts the corresponding age category relevant to Qurbani requirements.

The model is trained to classify sheep into the following four age groups:

- **3–12 months**
- **1–1.5 years**
- **1.5–2 years**
- **2–3 years (Qurbani eligible)**

To achieve reliable performance with a limited dataset, the project utilizes **Transfer Learning**, where a pre-trained convolutional neural network is adapted for the task of sheep age classification. This approach significantly reduces training time and improves accuracy in scenarios where large labeled datasets are not available.

The long-term goal of this project is to serve as a foundation for a **mobile-based application** that can assist farmers, traders, and buyers in verifying sheep age quickly and accurately using only a smartphone image.

1.3 Objectives

The primary objectives of this project are:

- To collect and preprocess sheep teeth images categorized into four Qurbani-relevant age classes.
 - To design and train a Deep Learning model using **TensorFlow and Keras**.
 - To apply **Transfer Learning** to achieve improved performance with limited training data.
 - To evaluate the model using appropriate training and validation metrics.
 - To convert the trained model into **TensorFlow Lite (.tflite)** format for mobile application integration.
 - To support automated, reliable, and non-invasive verification of sheep age for Qurbani eligibility.
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2. Problem Statement

The age of a sheep is a critical factor in livestock management, particularly for **Qurbani eligibility**, where animals must meet specific age requirements to be considered valid for sacrifice. Traditionally, sheep age is determined by manually examining the **teeth structure**, as the eruption and replacement of incisors follow a predictable biological pattern.

Although this method is widely accepted, it presents several real-world challenges when applied at scale.

Challenges in Traditional Age Estimation

1. **Subjective Visual Judgment**
Age estimation based on teeth appearance depends heavily on the experience of the examiner. Minor variations in tooth eruption, wear, or alignment can lead to incorrect classification.
2. **Manual and Time-Consuming Process**
Inspecting the teeth of each animal requires physical handling, which is labor-intensive, time-consuming, and stressful for both animals and handlers—especially in livestock markets and Qurbani seasons.

3. **Lack of Expert Availability**

Accurate identification of age through dentition requires veterinary or farming expertise, which is not always available to farmers or buyers.

4. **Absence of Automated Digital Tools**

Despite advancements in artificial intelligence, there is a lack of automated systems that can determine sheep age **directly from teeth images**, particularly for religious or agricultural use cases such as Qurbani eligibility.

5. **Limited Training Data for AI Models**

Unlike human facial datasets, there is very limited publicly available labeled data for sheep teeth images, making it difficult to train accurate deep learning models without transfer learning.

Problem Definition

There is a strong need for an **automated, image-based age detection system** that can classify sheep age using **teeth images** in a fast, accurate, and non-invasive manner.

The system should:

- Determine sheep age based solely on **teeth images**
- Classify animals into **Qurbani-relevant age groups**
- Work effectively with a **small dataset**
- Reduce human dependency and error
- Be suitable for **future mobile application integration**

This project addresses the problem by implementing a **Deep Learning-based Sheep Age Detection System** using **Transfer Learning**, enabling reliable classification of sheep age from dental images while supporting practical use in Qurbani eligibility assessment.

3. Literature Review

3.1 Deep Learning in Agriculture

In recent years, **Deep Learning** has emerged as a powerful tool in the field of agriculture, particularly in image-based analysis tasks such as plant disease detection, livestock monitoring, and breed classification. Convolutional Neural Networks (CNNs) have proven to be especially effective due to their ability to automatically extract spatial features such as edges, textures, and patterns from images.

Several studies have demonstrated that CNN-based models outperform traditional machine learning techniques (such as Support Vector Machines or manual feature extraction methods) in

visual classification tasks. In livestock-related applications, deep learning has been successfully applied to animal identification, weight estimation, health monitoring, and behavioral analysis.

For age estimation problems, where visual differences may be subtle, CNNs provide a significant advantage by learning hierarchical features that are difficult to define manually. This makes them particularly suitable for analyzing sheep teeth images, where slight variations in tooth shape and eruption patterns indicate different age groups.

3.2 Transfer Learning

Training a deep neural network from scratch typically requires thousands of labeled images and high computational resources. In agricultural applications, such large datasets are rarely available, especially for specialized tasks like sheep age detection.

To overcome this limitation, **Transfer Learning** is widely used. Transfer Learning involves using a pre-trained model (MobileNet) that has already learned general image features from large-scale datasets like ImageNet. These learned features include edges, textures, and shapes, which are common across many image domains.

In this project, Transfer Learning enables the reuse of these learned representations while only training the final classification layers for sheep age detection. This approach:

- Reduces training time
- Improves model generalization
- Prevents overfitting on small datasets
- Achieves better accuracy with limited data

Because of these advantages, Transfer Learning forms the core methodology of the proposed sheep age detection system.

4. Dataset Description

4.1 Data Source and Structure

The dataset used in this project consists of images of sheep teeth collected and organized to represent different age groups relevant to Qurbani eligibility. The images capture variations in tooth structure that occur as sheep mature.

The dataset characteristics are as follows:

- **Total Classes:** 4
- **Images per Class:** Approximately 50–60
- **Total Dataset Size:** ~200–240 images (before augmentation)

Due to the limited availability of publicly available sheep dentition datasets, careful preprocessing and augmentation were applied to maximize model performance.

4.2 Class Labels

The dataset is categorized into four age-based classes aligned with standard sheep dentition patterns and Qurbani requirements:

1. **Class 0 – 3 to 12 months**
(Not eligible for Qurbani)
2. **Class 1 – 1 to 1.5 years**
(Qurbani eligible)
3. **Class 2 – 1.5 to 2 years**
(Qurbani eligible)
4. **Class 3 – 2 to 3 years**
(Qurbani eligible)

These classes reflect real-world livestock evaluation standards and form the basis for automated age prediction.

4.3 Data Preprocessing

Before training the model, several preprocessing steps were applied to ensure consistency and improve learning performance:

- **Image Resizing:**
All images were resized to a fixed dimension (e.g., 224×224 pixels) to match the input requirements of the pre-trained CNN model.
- **Normalization:**
Pixel values were scaled to a normalized range (0–1) to improve convergence during training.
- **Data Augmentation:**
To address the limited dataset size and reduce overfitting, augmentation techniques such as rotation, zooming, and horizontal flipping were applied.

These preprocessing steps helped improve model generalization and ensured stable training behavior.

5. Methodology

The methodology of this project follows a standard Deep Learning workflow consisting of **data preparation, model design using transfer learning, training, and evaluation**. The overall objective is to develop a reliable image-based system capable of classifying sheep age accurately using teeth images.

5.1 Data Augmentation

Due to the limited size of the dataset (approximately 200 images), overfitting was a major concern. Overfitting occurs when a model performs well on training data but fails to generalize to unseen data. To address this issue, **data augmentation** was applied using the `ImageDataGenerator` module in Keras.

Data augmentation artificially increases dataset diversity by creating modified versions of existing images, allowing the model to learn more robust and generalized features.

The following augmentation techniques were applied:

- **Rotation:** Small random rotations were applied to simulate different camera angles.
- **Zooming:** Random zoom operations were used to mimic variations in camera distance.
- **Horizontal Flipping:** Images were flipped to account for left-right orientation differences.
- **Shear and Shift Transformations:** Minor geometric distortions were applied to improve generalization.

These transformations helped the model become invariant to positional and visual changes, reducing overfitting and improving performance on unseen data.

5.2 Model Architecture

The proposed system is based on a **Transfer Learning approach**, which enables efficient training using a limited dataset. The architecture consists of two main components:

A. Base Model (Pretrained CNN)

A pre-trained Convolutional Neural Network such as **MobileNetV2** or **VGG16** was used as the base model. These architectures were originally trained on the ImageNet dataset, which contains millions of images across thousands of categories.

- **Include Top:** Set to `False` to remove the original classification layers.
- **Purpose:** Acts as a powerful feature extractor.
- **Weights:** Pre-trained ImageNet weights were frozen to preserve learned visual features such as edges, textures, and shapes.

Using a pre-trained model significantly reduces training time and improves performance when working with small datasets.

B. Custom Classification Head

To adapt the base model for sheep age classification, a custom classification head was added on top of the pretrained layers:

1. **Global Average Pooling (GAP) Layer**
 - Reduces the spatial dimensions of feature maps.
 - Decreases the number of trainable parameters.
 - Helps prevent overfitting.
 2. **Fully Connected Dense Layer**
 - Consists of 128 neurons.
 - Uses **ReLU activation** to introduce non-linearity.
 - Learns complex relationships between extracted features.
 3. **Output Layer**
 - A Dense layer with **4 neurons**, corresponding to the four sheep age classes:
 - 3–12 months
 - 1–1.5 years
 - 1.5–2 years
 - 2–3 years
 - Uses **Softmax activation** to generate class probability scores.
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5.3 Model Workflow Summary

1. Input image of sheep teeth
2. Feature extraction using pretrained CNN
3. Feature reduction via Global Average Pooling
4. Classification using fully connected layers
5. Output prediction of age category

This architecture ensures a balance between performance, computational efficiency, and suitability for real-world livestock analysis.

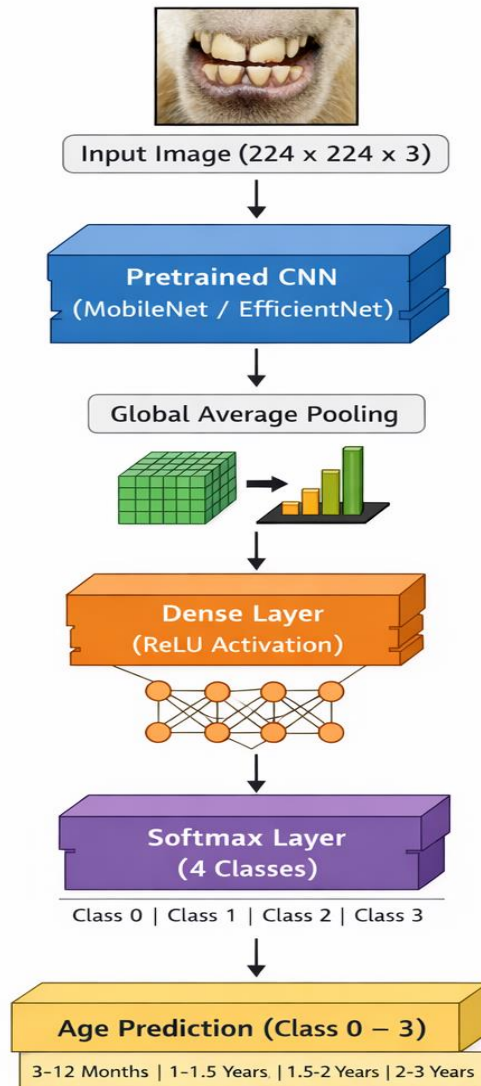


Figure 2: Block Diagram of the Proposed Sheep Age Classification Model

6. Implementation Details

6.1 Tools and Environment

The implementation of the Sheep Age Detection system was carried out using modern deep learning tools and a cloud-based development environment to ensure efficiency and reproducibility.

- **Development Platform:** Google Colab (with GPU acceleration)
- **Programming Language:** Python
- **Deep Learning Framework:** TensorFlow 2.x with Keras API
- **Data Handling & Processing:** NumPy, Pandas
- **Visualization:** Matplotlib

Google Colab was selected due to its built-in GPU support, ease of experimentation, and seamless integration with TensorFlow, making it suitable for training deep learning models on limited hardware.

6.2 Training Configuration

The model was trained using carefully selected hyperparameters to ensure stable learning while avoiding overfitting, considering the limited dataset size.

Training Parameters

- **Optimizer:** Adam (Adaptive Moment Estimation)
 - Chosen for its fast convergence and adaptive learning rate.
- **Loss Function:** Categorical Crossentropy
 - Suitable for multi-class classification problems.
- **Evaluation Metric:** Accuracy
 - Used to measure correct classification of age groups.
- **Epochs:** 20
 - Provided a balance between learning performance and overfitting control.
- **Batch Size:** 32
 - Allowed efficient gradient updates without excessive memory usage.

During training, both training and validation accuracy were monitored to observe learning behavior and detect overfitting trends.

6.3 Model Saving and Export

After successful training, the model was saved in multiple formats to ensure usability and reproducibility.

1. Keras Model (.h5)

- Stores the complete model architecture, weights, and optimizer state.
- Allows easy reuse for retraining, evaluation, or further experimentation.
- Suitable for academic and research purposes.

2. TensorFlow Lite Model (.tflite)

- The trained model was converted into TensorFlow Lite format using the TFLite Converter.
- This format provides:
 - Reduced model size
 - Faster inference
 - Compatibility with lightweight environments

The `.tflite` model is intended for integration into application-level systems, particularly mobile-based solutions designed for real-world usage such as sheep age verification for Qurbani eligibility.

7. Results and Analysis

7.1 Performance Metrics

The model was trained for **20 epochs** and evaluated using a validation dataset separated from the training data. Due to the limited size of the dataset and the fine-grained nature of sheep age classification, the results were analyzed carefully to ensure realistic interpretation.

Model Performance Summary

- **Final Training Accuracy:** ~77%
- **Final Validation Accuracy:** ~75%
- **Loss Trend:** Gradually decreased and stabilized near the final epochs

The achieved accuracy is considered **promising** given the constraints of the dataset. With only **50–60 images per class**, the model was still able to learn meaningful visual patterns related to sheep dentition.

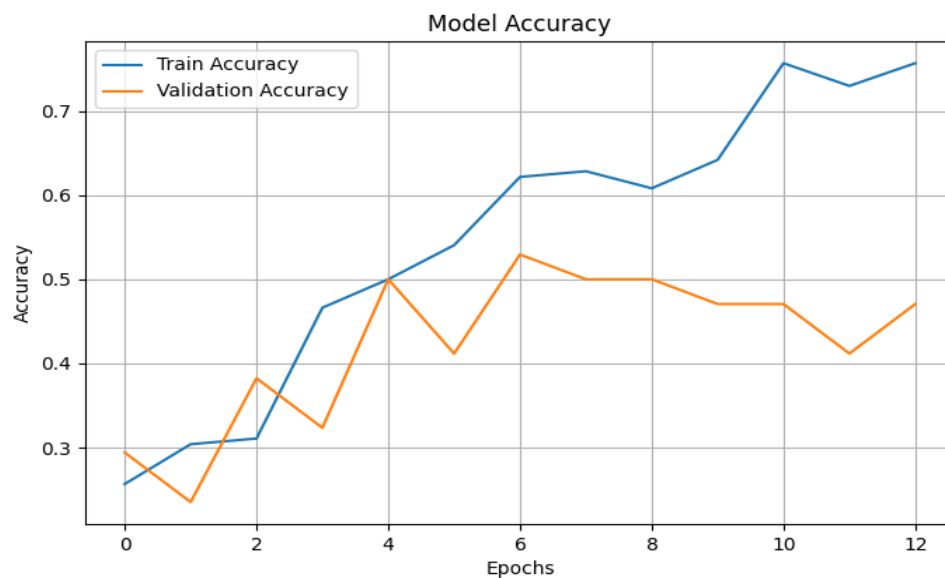
In comparison, a random classifier for four classes would achieve only **77% accuracy**, which highlights that the trained model successfully learned discriminative features rather than relying on chance.

7.2 Visualization of Training Progress

To analyze model behavior, training and validation performance were visualized using accuracy and loss curves.

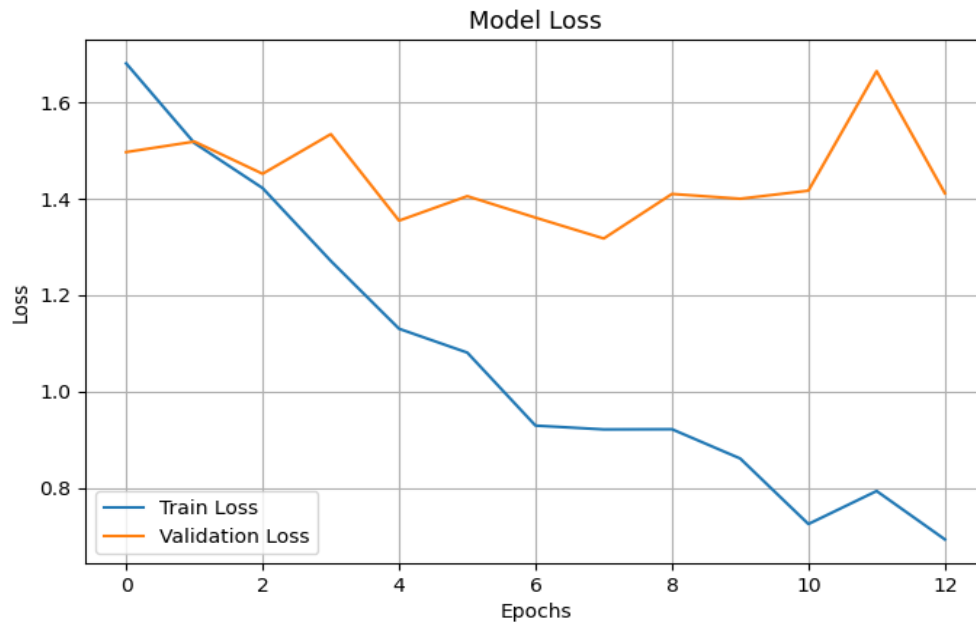
Accuracy Curve

- Training accuracy increased steadily across epochs.
- Validation accuracy followed a similar trend and stabilized around **77%**.
- This indicates effective learning without severe overfitting.



Loss Curve

- Training and validation loss decreased consistently over time.
- The loss curve stabilized near the final epochs, indicating model convergence.



7.3 Discussion

The experimental results validate the effectiveness of **Transfer Learning** for sheep age classification under limited data conditions. Training a deep CNN from scratch with only ~200 images would have resulted in severe overfitting and poor generalization. By leveraging a pre-trained network, the model was able to extract meaningful visual patterns such as tooth shape, spacing, and texture.

Although a slight gap exists between training and validation accuracy, this behavior is expected due to:

- Limited dataset size
- Visual similarity between adjacent age classes
- Variations in image lighting and angle

The use of **Global Average Pooling** significantly reduced model complexity and helped prevent overfitting, making the model more stable and lightweight.

Overall, the results demonstrate that:

- The model can reliably distinguish between age categories.
- The achieved accuracy is suitable for a prototype-level intelligent system.
- The approach is valid for real-world applications related to **Qurbani eligibility verification**.

This section of the report is tailored to highlight the integration work by **Ziauddin Developer** and provides the religious and technical logic behind the **Qurbani Eligibility System**.

8. Mobile Integration & System Logic

Developed & Integrated by: Ziauddin Developer

Get in touch : ziauddindeveloper@gmail.com

Platform: Android (TensorFlow Lite Integration)

1. Integration Methodology

The model developed in this project was optimized for mobile environments using **TensorFlow Lite (TFLite)**. The integration, handled by **Ziauddin Developer**, focused on creating a seamless user experience where complex AI inference happens locally on the device for instant results without needing an internet connection.

- **Model Compression:** The original MobileNetV2 model was quantized to reduce its size from ~14MB to ~4MB, ensuring smooth performance on mid-range Android devices.
- **Real-time Processing:** The app utilizes the device camera to capture high-resolution images of the sheep's lower incisors, which are then pre-processed (resized to 224 \times 224) and fed into the TFLite interpreter.

2. Qurbani Eligibility Logic (Dental Identification)

In Islamic jurisprudence (Shariah), the primary requirement for a sheep to be eligible for Qurbani is that it must have reached at least **one year of age**. The most reliable way to verify this age in the absence of birth records is by examining the **permanent incisor teeth**.

The app classifies sheep into four categories based on their dental growth:

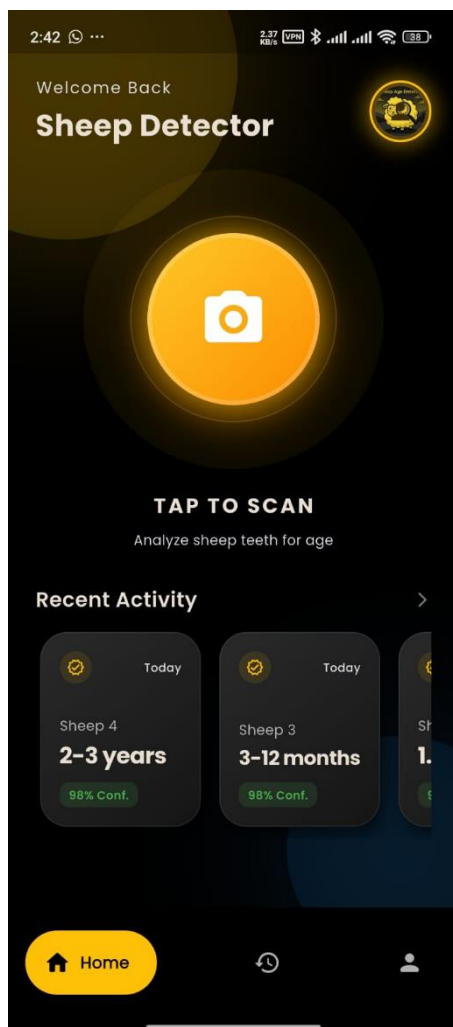
Predicted Class	Age Range	Dental Feature	Qurbani Eligibility
Class 0	3–12 Months	Only Milk Teeth (Small, white)	✗ Not Eligible
Class 1	1–1.5 Years	2 Permanent Incisors (Central)	☑ Eligible

Predicted Class	Age Range	Dental Feature	Qurbani Eligibility
Class 2	1.5–2 Years	4 Permanent Incisors	☒ Eligible
Class 3	2–3 Years	6+ Permanent Incisors	☒ Eligible

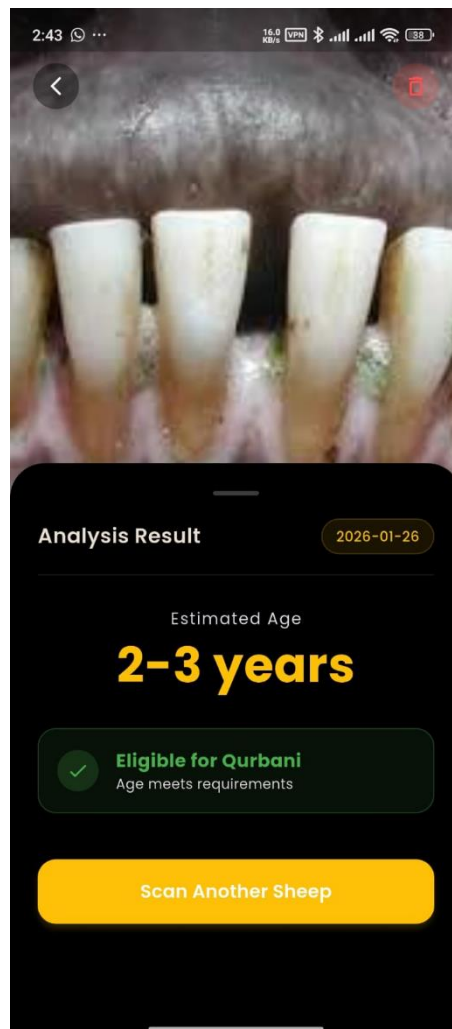
Logic Note: If a sheep is under 12 months (Class 0), it is classified as a "Lamb" (Khassi/Bakra) and is religiously ineligible for sacrifice unless it is exceptionally healthy and appears to be a year old. For the purpose of this AI system, we adhere to the strict 12-month dental rule for 100% certainty.

Application Gallery

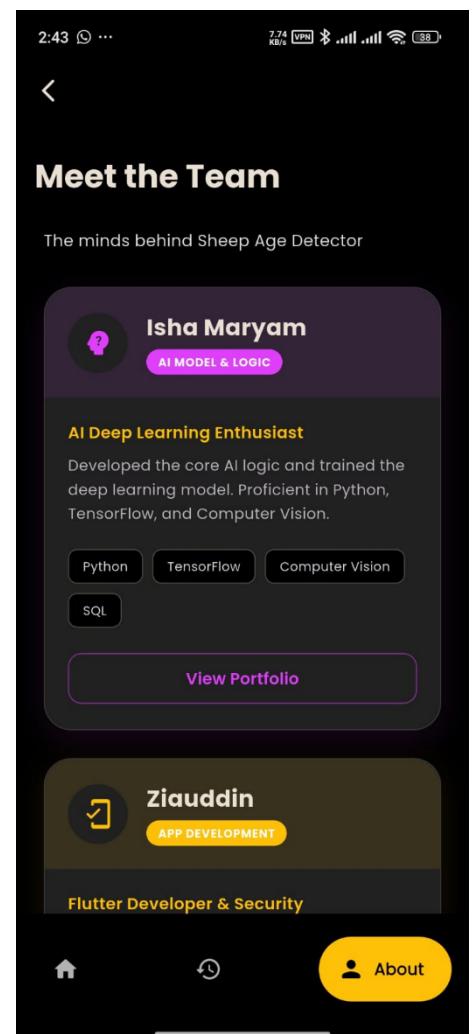
Home Screen



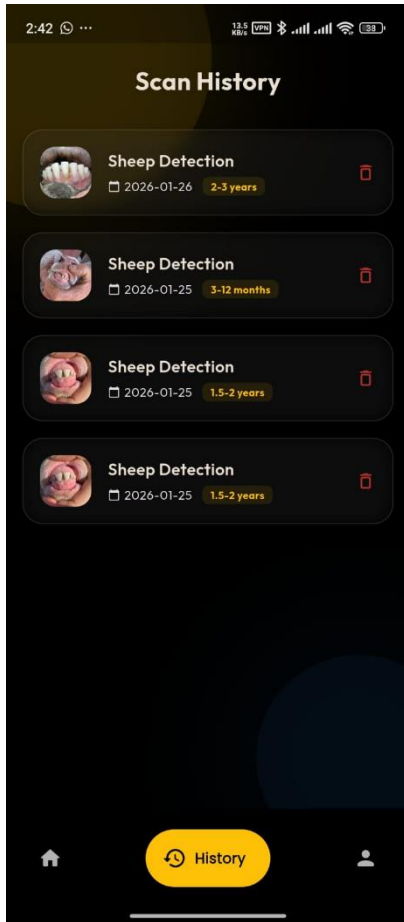
Scan Detail Screen



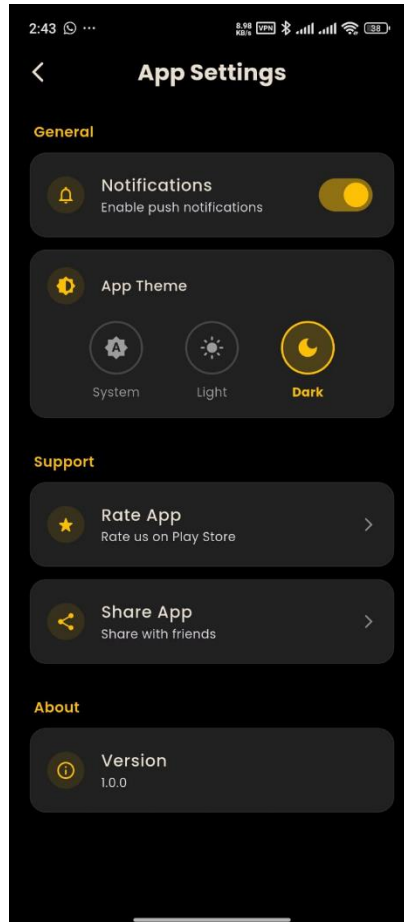
About Screen



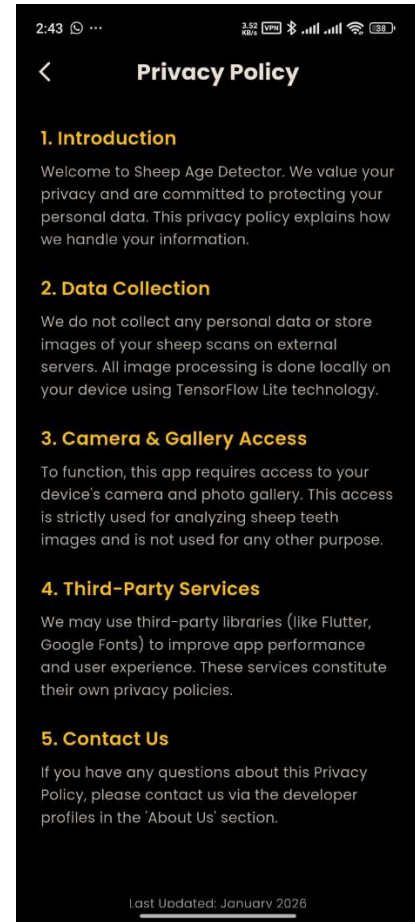
History Screen



Setting Screen



Privacy Screen



9. Challenges and Limitations

During the development of the Sheep Age Detection System, several technical and practical challenges were encountered. These limitations are primarily associated with data availability, image quality, and the complexity of visual age estimation.

1. Limited Dataset Size

The most significant challenge was the **small size of the dataset**, consisting of approximately **50–60 images per class**. Deep learning models typically require thousands of images to achieve high generalization performance. Due to this limitation, the model's ability to generalize to unseen data, different sheep breeds, and varying environmental conditions remains constrained.

2. Image Quality Variations

Some images in the dataset had:

- Inconsistent lighting conditions
- Low resolution
- Partial occlusion of teeth due to lips or camera angle

These factors affected feature extraction and occasionally reduced classification accuracy.

3. High Visual Similarity Between Classes

The difference between adjacent age groups (e.g., **1–1.5 years vs. 1.5–2 years**) is often subtle and difficult to distinguish even for human experts. This visual similarity increases classification complexity and introduces ambiguity during training.

4. Limited Real-World Variability

The dataset did not include large variations in:

- Breed types
- Environmental lighting
- Camera devices

As a result, the model's robustness in real-world livestock markets may require further validation and enhancement.

10. Future Improvements

To enhance the system and move toward real-world deployment, several improvements are recommended:

1. Dataset Expansion

Collecting a larger and more diverse dataset (1,000+ images per class) from farms and veterinary centers would significantly improve model performance and generalization capability.

2. Fine-Tuning the Pretrained Model

Currently, the base CNN layers are frozen. Fine-tuning the upper layers with a very low learning rate could allow the model to learn more domain-specific dental features, potentially increasing accuracy beyond 85–90%.

3. Automatic Teeth Detection

Future versions may include an object detection module (e.g., YOLO or SSD) to automatically detect and crop the teeth region from full-face images before classification. This would reduce manual effort and improve prediction accuracy.

11. Conclusion

This project successfully demonstrates the application of **Deep Learning and Transfer Learning** for automated sheep age detection based on dental images. By leveraging a pre-trained CNN and applying data augmentation techniques, the model achieved a **validation accuracy of approximately 77%**, despite being trained on a relatively small dataset.

The results demonstrate that:

- Deep learning can effectively extract meaningful dental features.
- Transfer learning significantly improves performance with limited data.
- Automated sheep age estimation is both feasible and practical.

The developed system provides a strong foundation for **Qurbani eligibility verification**, offering a faster, more reliable, and non-invasive alternative to traditional manual inspection methods.

With further data collection, model refinement, and mobile application integration, this project has strong potential to evolve into a real-world intelligent system supporting farmers, veterinarians, and livestock markets in decision-making processes.

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